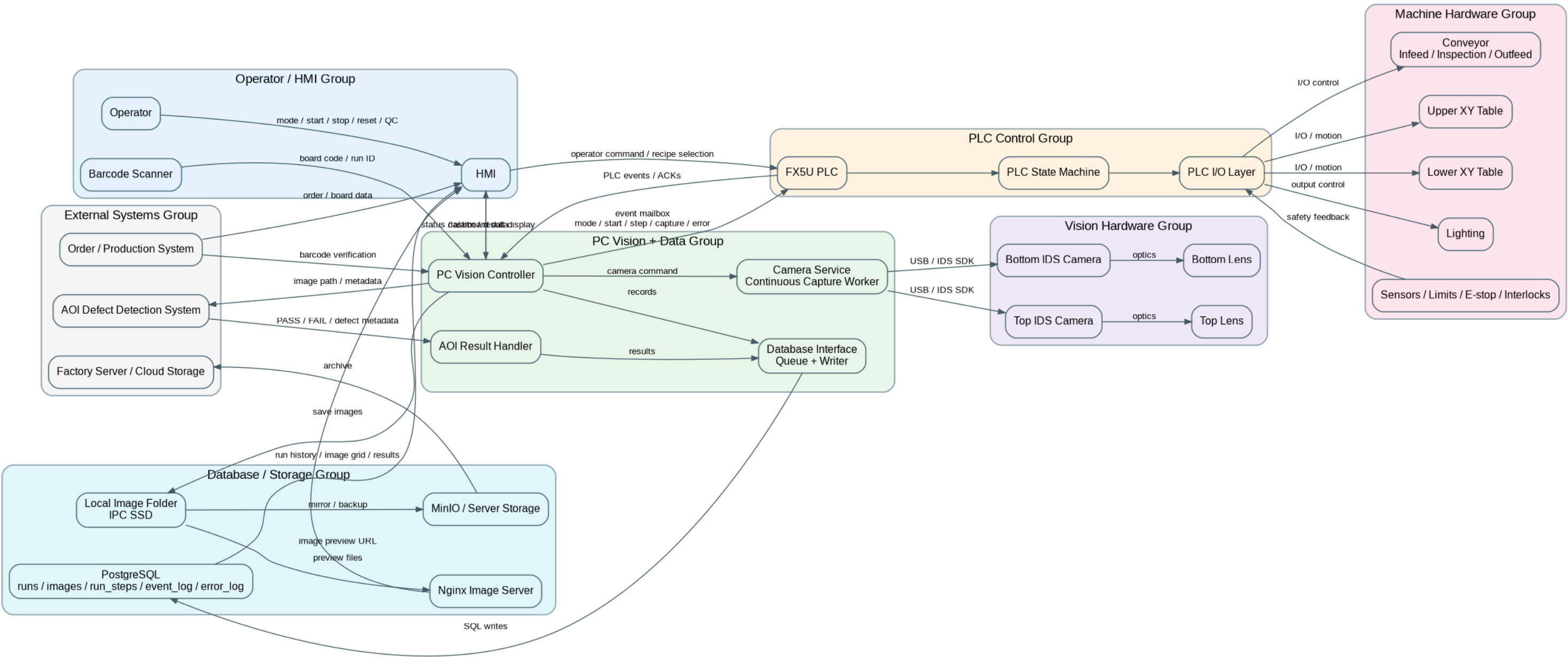


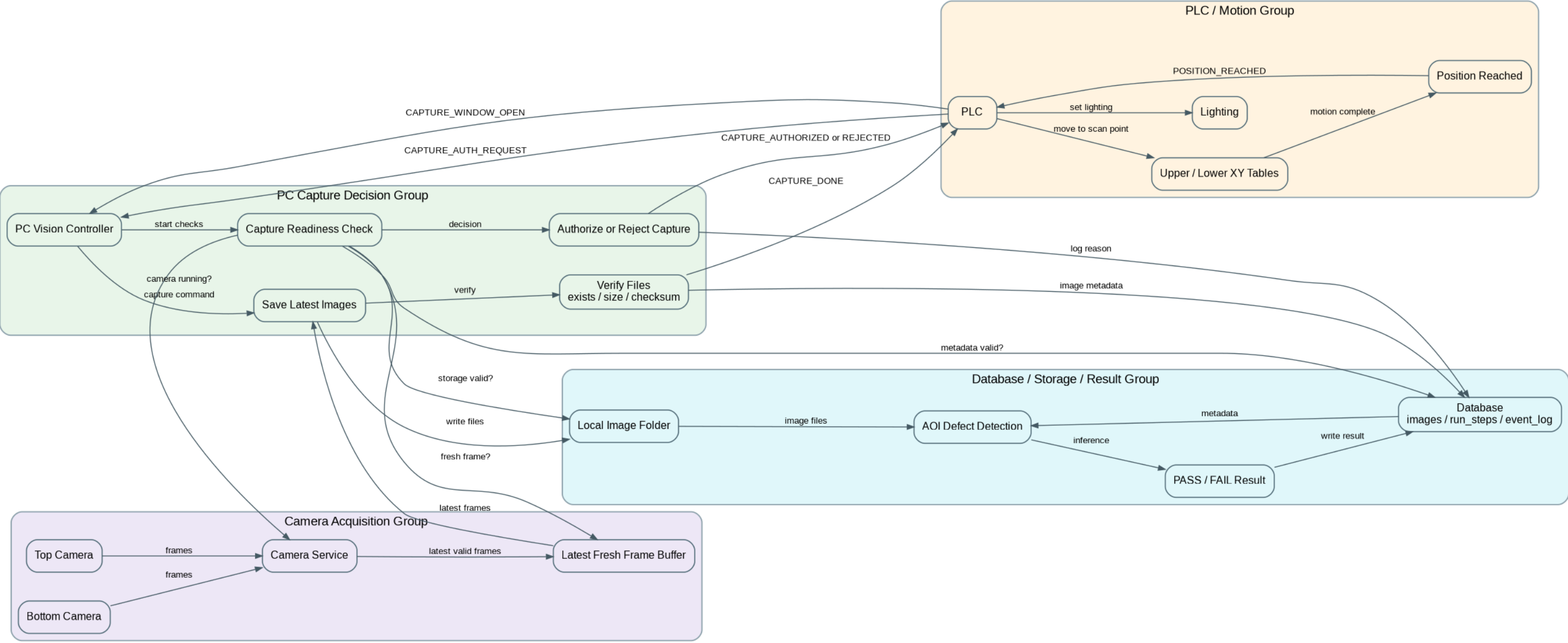
Overall Functional Connection

DUO_AOI Overall Functional Connection Diagram



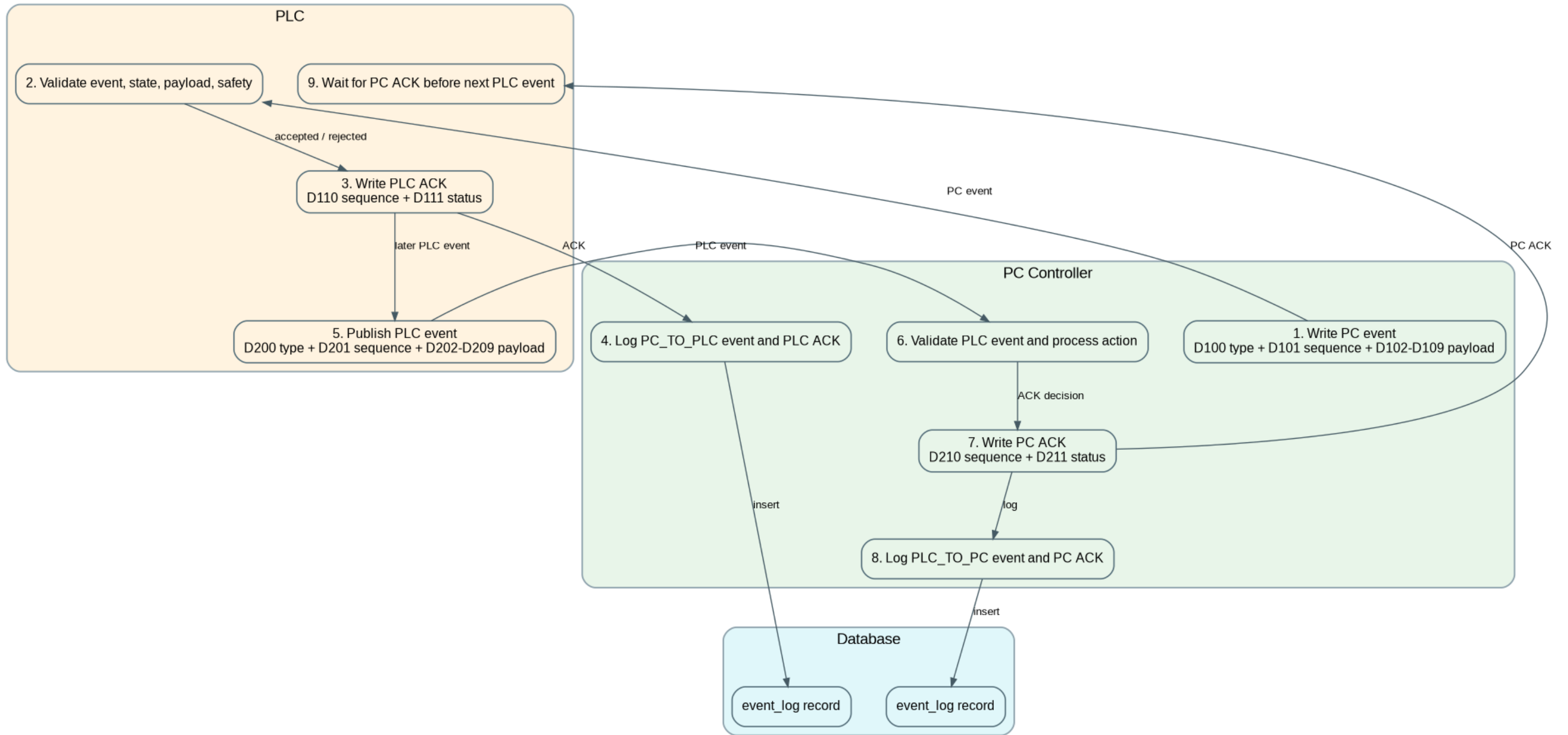
Vision Capture Dataflow

Vision Capture Dataflow Diagram



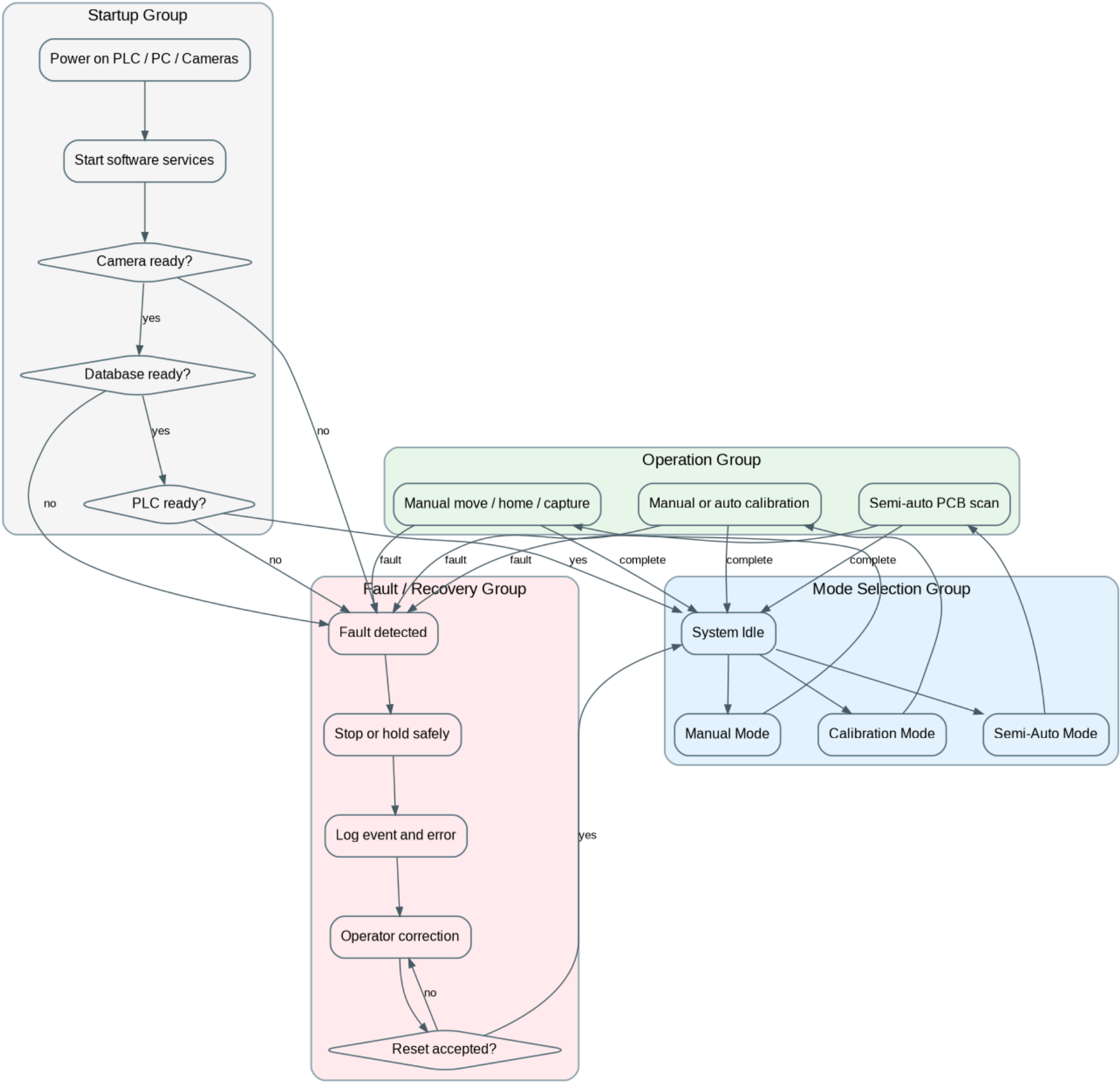
Pc Plc Event Communication Sequence

PC / PLC Event Communication Sequence

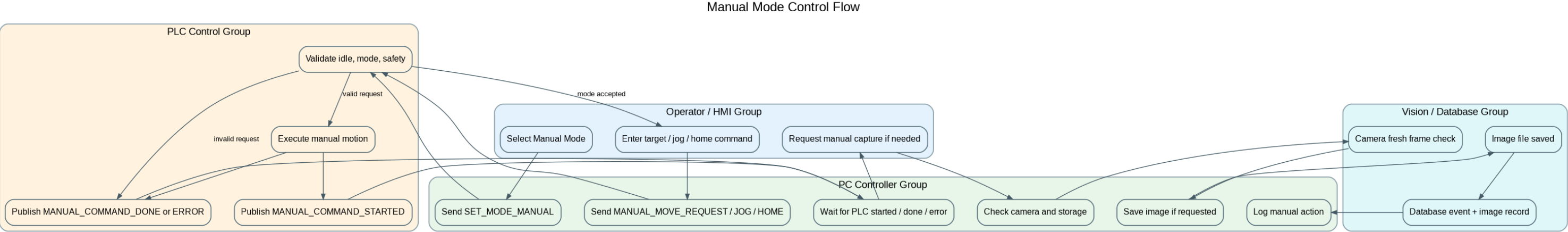


High Level Control Flow

High-Level Control Flow Graph

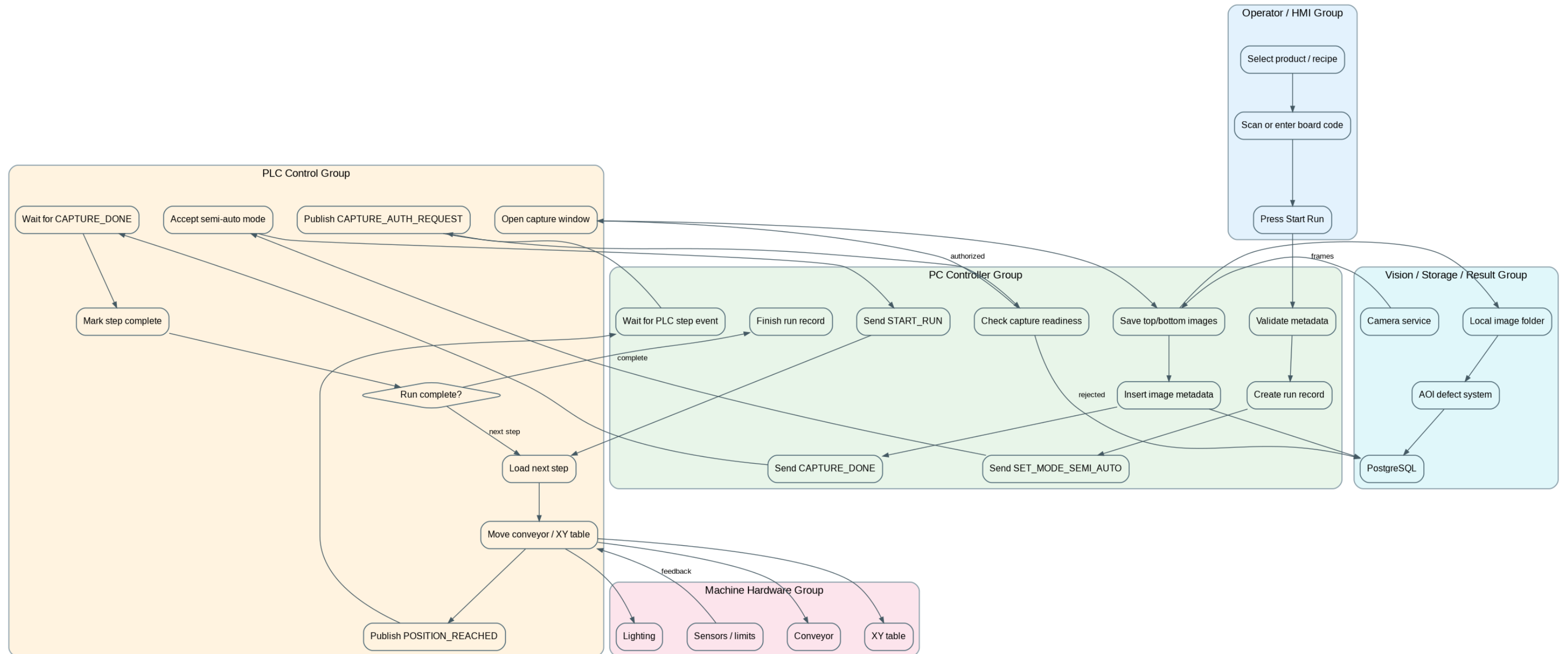


Manual Mode Control Flow



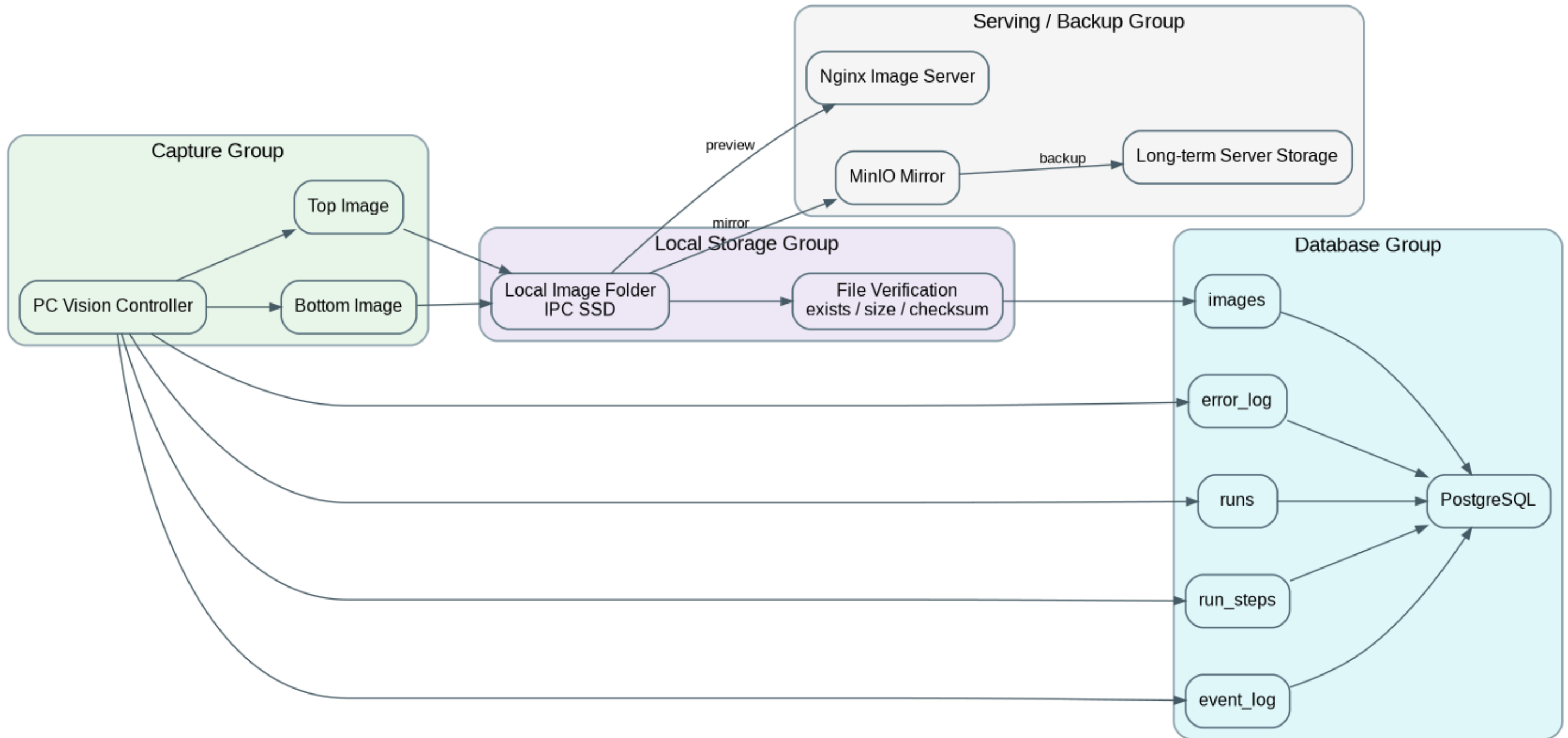
Semi Auto Mode Control Flow

Semi-Auto Mode Control Flow

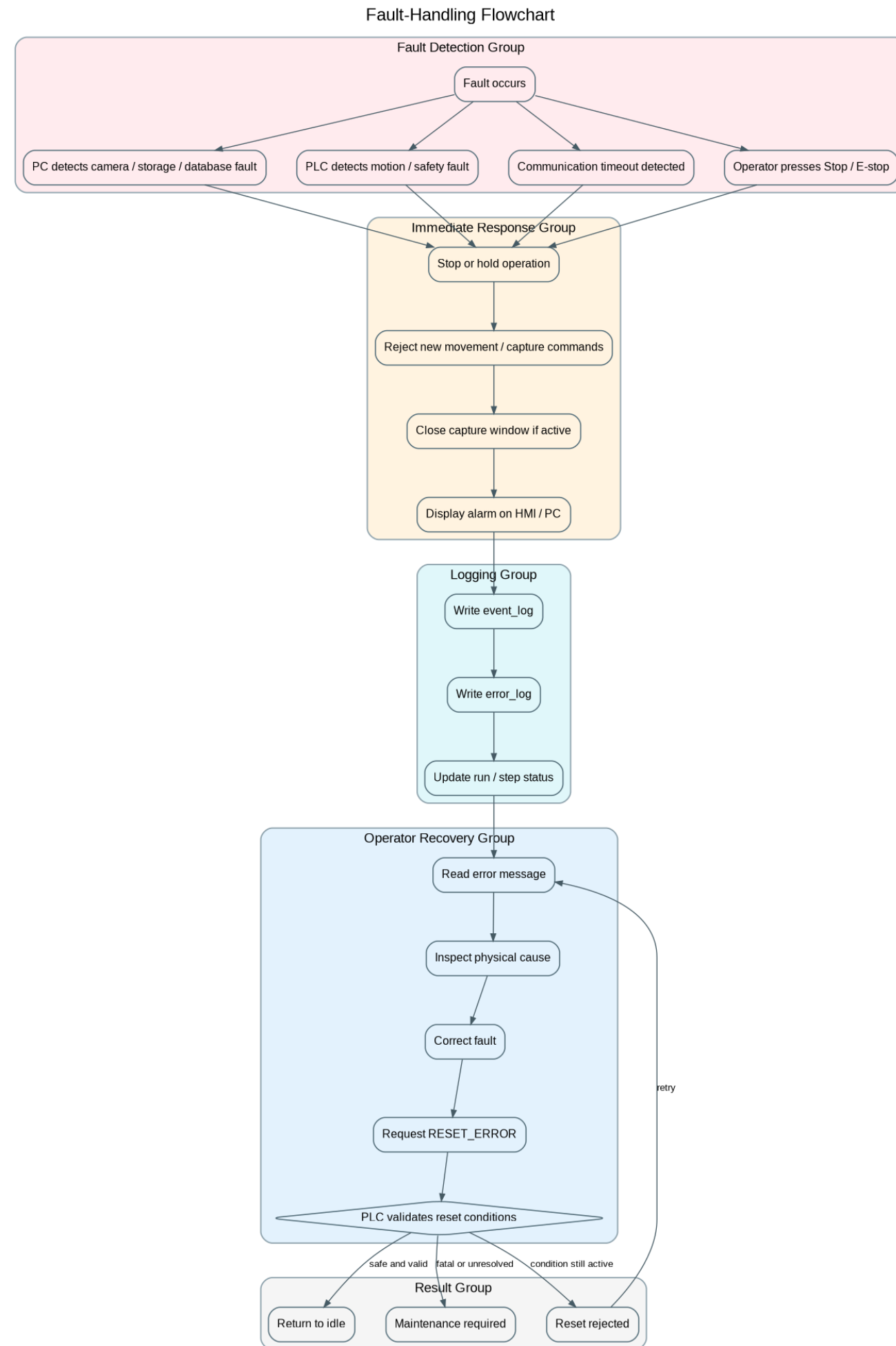


Database Storage Flow

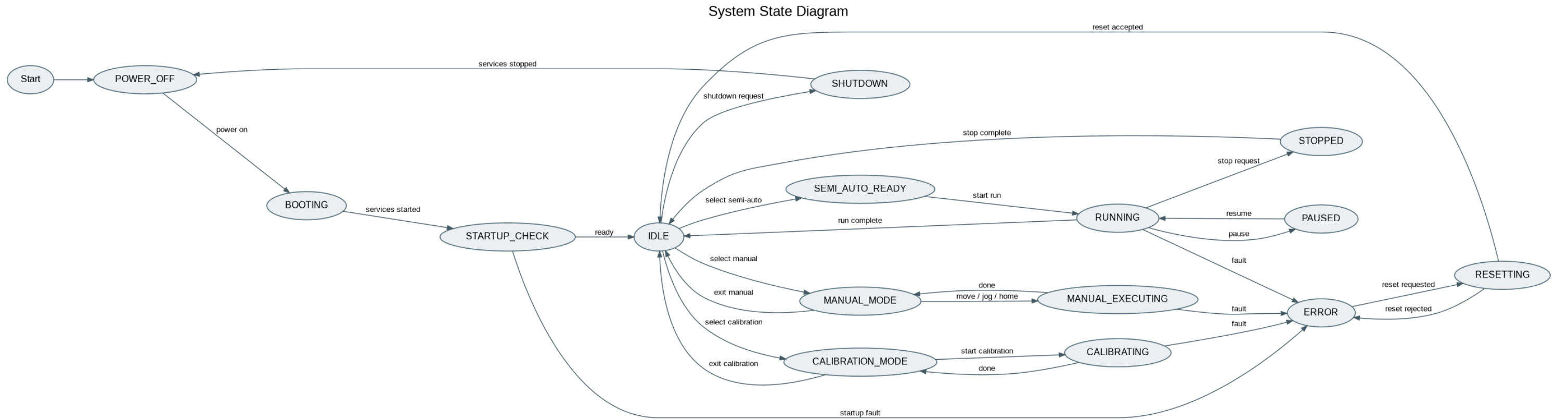
Database and Storage Flow



Fault Handling Flowchart

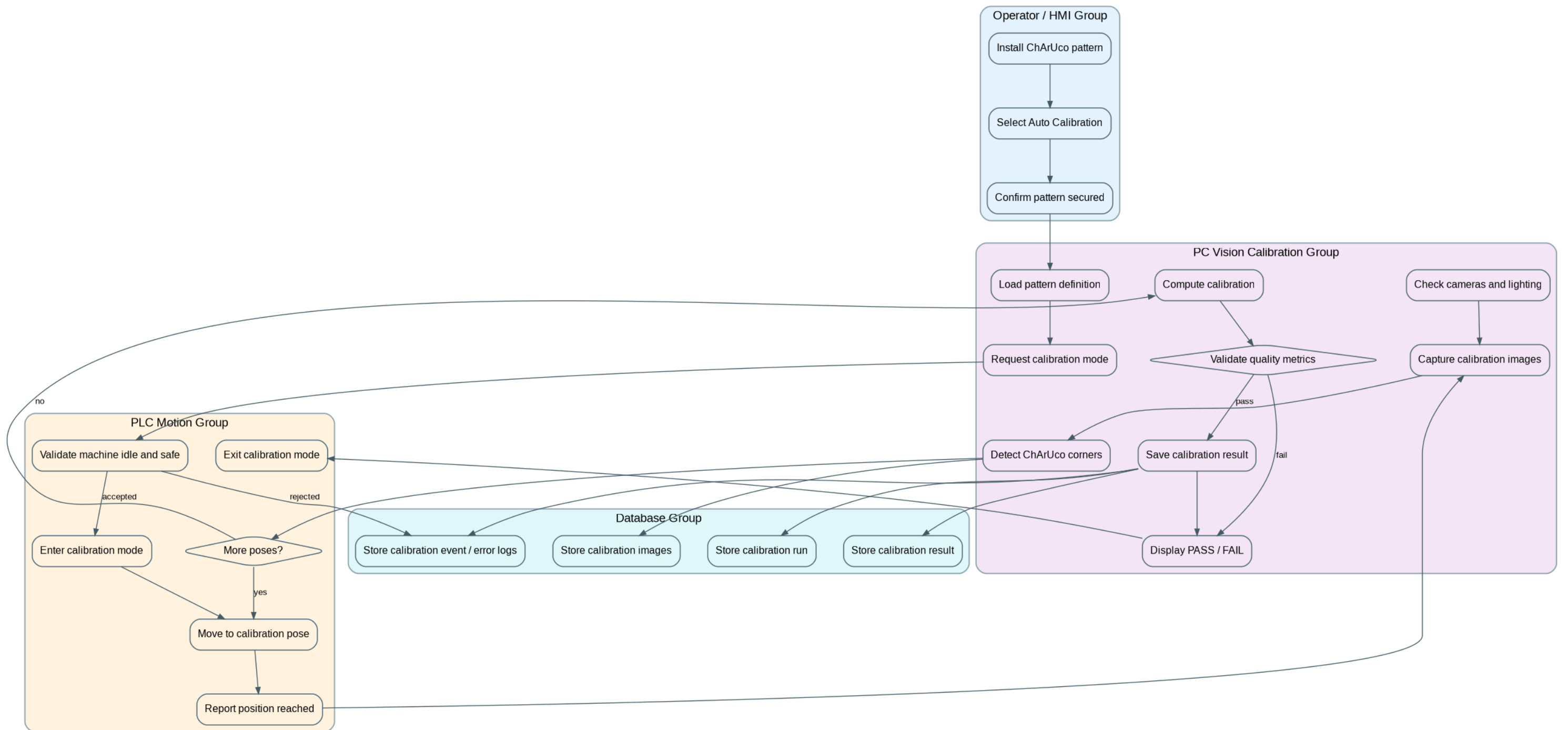


System State Diagram

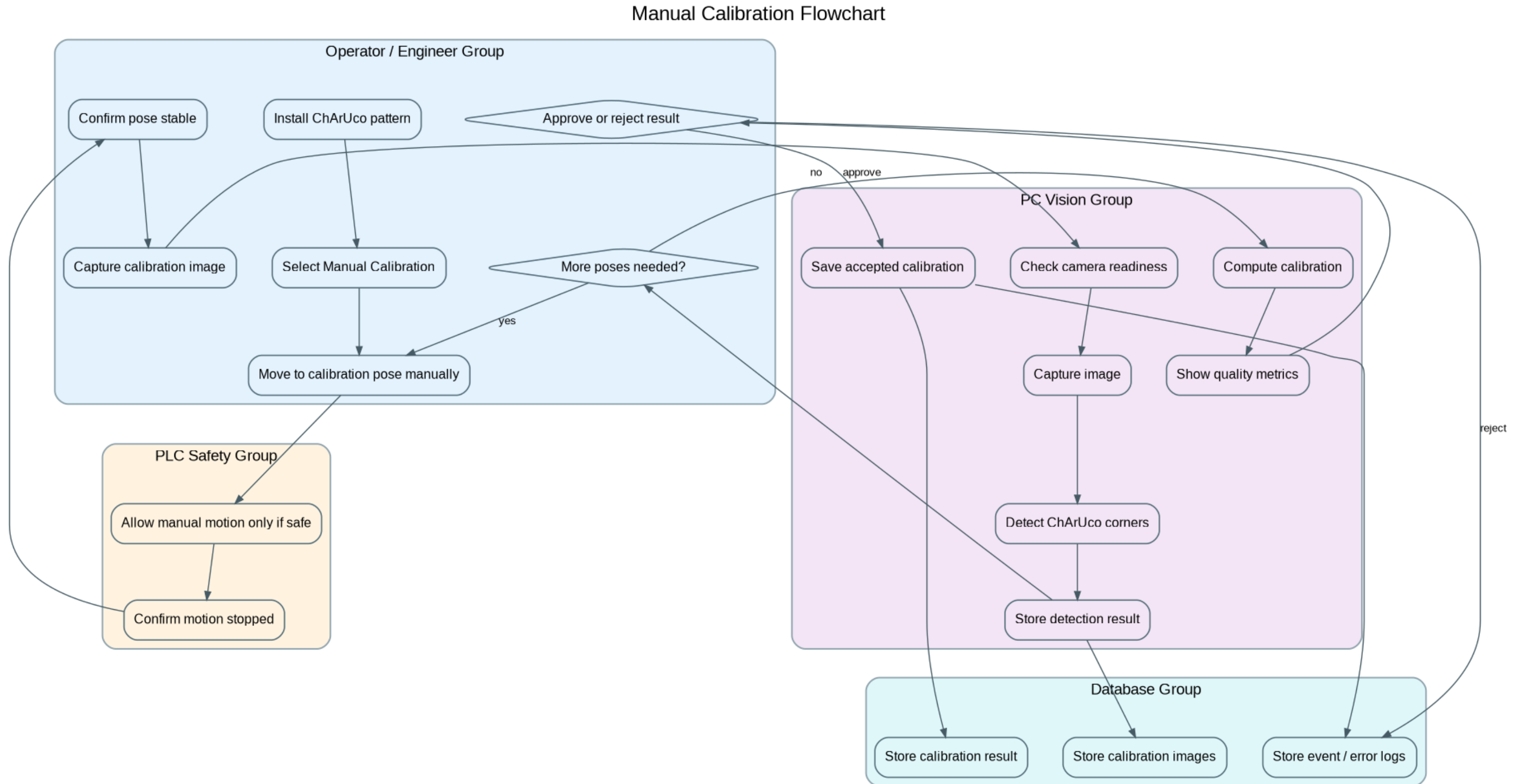


Auto Calibration Flowchart

Auto Calibration Flowchart



Manual Calibration Flowchart



Operating Sequence Error Reverse Chart

Operating Sequence to Error-Code Reverse Chart

Operating Step	Possible Fault	Error Range	Example Code	Required Response
Software startup	PC configuration missing	1000	1004	Load correct configuration
PLC connection	PLC not reachable	3000	3001	Check Ethernet, IP, port, PLC state
Camera startup	Camera not detected	4000	4001	Check USB cable, camera power, SDK
Database startup	Database unavailable	5000	5010	Check PostgreSQL, Docker, fallback logging
Mode selection	Mode switch while busy	1000/2000	1010/2002	Wait until machine is idle
Manual move	Axis not homed	6000	6001	Run homing sequence
Manual move	Target out of range	6000	6003	Correct target position
Semi-auto start	Metadata missing	8000	8001/8002	Enter board code and side
Semi-auto start	Recipe missing	2000	2012	Load correct recipe
Move to scan point	Motion timeout	6000	6006	Inspect obstruction, drive, target
Position reached	PC/PLC step mismatch	1000/6000	1008/6013	Stop and inspect event log
Capture authorization	Camera stale frame	4000	4011	Restart or verify acquisition
Capture authorization	Storage path invalid	5000	5002	Correct save folder
Capture window	PC does not respond	3000	3010	Check PC software and communication
Image save	File save failed	5000	5006	Check disk, folder, permissions
Image verification	File invalid or checksum failed	5000	5007	Re-capture image
Any operation	E-stop active	7000	7001	Inspect machine before reset
Any operation	Heartbeat lost	3000	3011	Safe stop, check network and software

Test Matrix Chart

Test Matrix Chart

Test Group	Test Case	Expected Result
Startup	PLC power on	PLC reaches ready state
Startup	PC software launch	PC reaches ready state
Startup	Camera connection	Top and bottom cameras detected
Startup	Database connection	Database connected
Communication	PC sends PC_READY	PLC replies PLC_READY
Communication	PC requests manual mode	PLC ACKs and publishes MODE_CHANGED
Communication	PC requests semi-auto mode	PLC ACKs and publishes MODE_CHANGED
Communication	PLC publishes event	PC ACKs before next PLC event
Communication	Heartbeat active	No timeout alarm
Vision	Continuous acquisition	Frames update continuously
Vision	Fresh frame check	Valid frame timestamp detected
Vision	Manual capture	Image saved and verified
Vision	Dual-camera capture	Top and bottom images saved
Database	Image insert	images row created
Database	Event insert	event_log row created
Database	Error insert	error_log row created
Control	Manual home	PLC homes safely
Control	Manual move	PLC moves and reports done
Control	Out-of-range move	PLC rejects command
Control	Semi-auto one step	Move, capture, and step complete
Fault injection	Camera disconnected	PC rejects capture and logs camera error
Fault injection	Storage full	PC rejects capture and logs storage error
Fault injection	E-stop pressed	PLC enters safety error state
Fault injection	Database unavailable	PC blocks semi-auto start or uses fallback policy
Calibration	Auto calibration mode	PLC accepts only when idle and safe
Calibration	ChArUco detection	Required corners detected
Calibration	Calibration computation	Calibration result generated
Calibration	Calibration quality check	PASS/FAIL with metrics
Calibration	Calibration database save	Result and images stored